

Easy Visa

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Yashpal Singh

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Executive Summary –ML Driven Visa Predictability Solution

- This project successfully addressed the critical challenge facing the U.S. Office of Foreign Labor Certification (OFLC): managing a high volume of work visa applications (775,979 in FY 2016) that render manual review unsustainable, leading to delays and potential inefficiencies
- The objective was to develop a **Machine Learning–based classification model** to predict the likelihood of visa approval (**Certified vs. Denied**), thereby enabling a faster, data-driven, and scalable decision process
- The final solution is a **Gradient Boosting Classifier tuned with Oversampled Data**, selected for its superior performance and exceptional generalization capability on unseen data. The **F1-Score** was chosen as the primary metric, as it optimally balances **Precision** (minimizing false approvals) with **Recall** (maximizing true approvals), directly supporting the OFLC's need for both efficiency and compliance.

Executive Summary –ML Driven Visa Predictability Solution

Metric	Performance on Unseen Test Data	Key Takeaway
F1-Score	0.8282(82.8%)	Confirms robust and balanced predictive power
Recall	0.8728(87.8%)	Highly effective at identifying true positive visa approvals
Precession	0.7880(78.8%)	Strong confidence that predicted approvals will be correctly certified
Generalization	Validation F1-Score(0.8141) Training F1-Score(0.8133)	Zero overfitting , ensuring reliable performance upon deployment

Executive Summary –Key Actionable Insights & Recommendations

The ML model provides not only a prediction but also transparency by highlighting the most influential features, which forms the basis for strategic business and policy recommendations.

Strategic OFLC Process Streamlining

- **Recommendation:** Implement the Gradient Boosting model as an automated triage system for preliminary application screening.
- **Actionable Insight:** Applications predicted with a high probability of certification can be fast-tracked, allowing human reviewers to re-allocate up to 87.3% of their time to focus exclusively on complex, low-confidence, or borderline cases. **This significantly reduces the manual review burden and accelerates decision timelines**

Targeted Applicant/Employer Guidance (EasyVisa)

Recommendation: Leverage feature importance and EDA insights to provide data-driven consulting to employers and applicants, maximizing their application success rates.

Executive Summary –Key Actionable Insights & Recommendations

Key Predictor	Actionable Insight	Business Recommendations
Prior Job Experience	Applicants with prior job experience have a 20-percentage-point higher certification rate (approx. 75% vs. 55%).	Prioritize and highlight detailed documentation of prior job experience in applicant profiles
Unit of wage	Wages reported on an Annual basis have a significantly higher certification rate (~70%) compared to an Hourly basis (~35%)	Advise employers to standardize wage reporting to an Annual unit where feasible, to optimize the application's perceived stability and success rate.
Education & Location	The High School category is the single most dominant predictive factor in the model, though higher degrees (Master's, Doctorate) showed low relative importance. Geographic factors like Region of Employment (West, South, Northeast) and Continent of Origin are critical drivers	Focus on complete and accurate capture of all categorical metadata, as specific categories (like High School) are strong indicators, and geographic variables outweigh even the prevailing wage in predictive power

Business Problem

Business Context :

In today's competitive business environment, the demand for skilled human resources in the United States continues to grow. Companies increasingly rely on both domestic and foreign workers to fill critical workforce gaps. To regulate this process, the U.S. Department of Labor's **Office of Foreign Labor Certification (OFLC)** evaluates hundreds of thousands of employer applications each year to determine whether foreign workers can be granted work visas.

However, the **visa certification process is highly time-consuming and resource-intensive**. In FY 2016 alone, OFLC processed **775,979 applications for nearly 1.7 million positions**, reflecting a **9% year-over-year increase**. As the application volume continues to rise, manual review becomes unsustainable, leading to delays, increased administrative costs, and potential inefficiencies in decision-making.

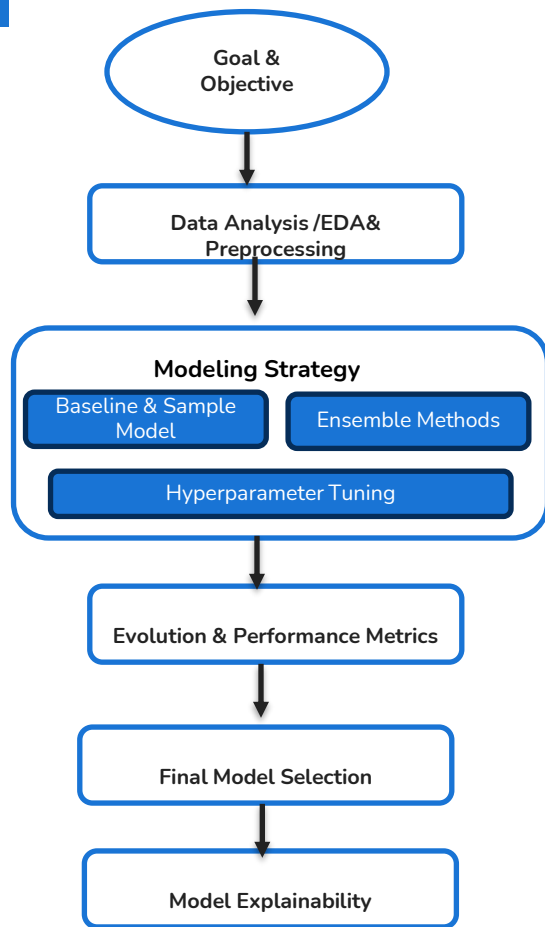
The critical challenge lies in **identifying which applications are more likely to result in visa approval** while ensuring compliance with statutory requirements and safeguarding U.S. workers' interests.

Objective

Being a Data Scientist at EasyVisa, analyze the data provided and to develop a **Machine Learning-based classification model** that can:

- **Predict the likelihood of visa approval** based on applicant and employer features.
- **Identify the key drivers** influencing approval decisions to provide transparency in recommendations.
- **Facilitate a faster and data-driven decision-making process**, enabling OFLC and employers to streamline workforce planning and **reduce** processing bottlenecks.
- By leveraging historical application data, this project seeks to empower EasyVisa to provide **OFLC with an intelligent system that supports efficient, fair, and scalable visa application assessments**.

Solution Approach - Strategy & Execution



Project Goal & ML Objective:

Goal :Develop an ML-based system for the OFLC to streamline the work visa application approval process.

Objective: Build a robust Classification Model to predict the binary outcome: Visa Approval (**Certified**) vs. **Denial** (Denied)

Data Analysis and Preprocessing :

Data Analysis:

- Identified the target variable (**CASE_STATUS**) and selected relevant predictor features for model development.
- Performed data cleaning by handling missing values (imputation/removal), correcting data types, and addressing outliers to ensure dataset consistency and reliability.

EDA:

- Conduct **univariate and bivariate analysis** to understand feature distributions and relationships with the target variable.
- Detect potential outliers and data skewness using visualization techniques.
- Generate a **correlation heatmap** to analyze inter-feature relationships and their influence on the target variable.

Feature Engineering:

- Develop new derived features to enhance model predictability and capture complex patterns.
- Apply categorical encoding (**One-Hot Encoding / Label Encoding**) for non-numeric attributes such as CASE_STATUS and other categorical fields, perform feature scaling and normalization to maintain consistency across features for model training.

Modeling Strategy/Comprehensive Classification:

1: Baseline & Simple Models: Start with **Logistic Regression** and **Decision Tree** for quick, interpretable baselines.

2: Ensemble Methods (The Core): Implement **Random Forest (RF)**, **Adaboost**, **Gradient Boosting Machines (GBM)**, **XGBoost** for superior predictive power, as they handle complex non-linear relationships well.

3: Hyperparameter Tuning: Systematically optimize the best-performing models (RF, GBM) using techniques like **Grid Search** and **Randomized Search** with **Cross-Validation (k-fold)** to find the optimal parameter combinations.

Performance Metric Evaluations

Primary Metric: F1-Score. Why? Visa approvals often represent an imbalanced class distribution. The F1-Score (Harmonic Mean of Precision and Recall) is the ideal metric for classification on imbalanced datasets, ensuring a good balance between minimizing False Positives (Incorrectly approving a denial) and False Negatives (Incorrectly denying an approval)

Model Explainability

Use Feature Importance plots to identify and rank the Key Drivers influencing the prediction of visa approval. This meets the transparency requirement

Solution Approach – Model Building and Selection

Data Split :

Split data into **Train** , **Validation** and **Test sets**. Stratify the split based on the target variable to maintain the same class ratio in both subsets

Model Evaluation Criterion:

- Choose the primary metric to evaluate the model on
- Create functions to calculate different metrics and confusion matrix so that we don't have to use the same code repeatedly for each model.
- Defining scorer to be used for cross-validation and hyperparameter tuning

Initial model training with original data:

- Train, Decision Tree, Bagging, Random Forest, Adaboost, GBM and XGBoost on the training data.
- **Performance Primary Metric comparison** with Training and Validation Set of the data

Model training with Oversampled data:

- Apply **SMOTE Technique** to make balance set of the classes on trained data set and get oversample trained dataset
- **Train all models**(Decision Tree, Bagging, Random Forest, Adaboost, GBM and XGBoost) with the oversampled of the trained DataSet
- **Performance Primary Metric comparison** with oversampled training and validation DataSets

Model training with Undersampled data:

- Apply **RandomUnderSampler Technique** to make balance set of the classes on trained data set and get Undersample trained dataset
- **Train all models**(Decision Tree, Bagging, Random Forest, Adaboost, GBM and XGBoost) with the Undersampled of the trained DataSet
- **Performance Primary Metric comparison** with oversampled training and validation DataSets

Solution Approach – Model Building and Selection

1. AdaBoost Tuning with Oversampled Data

- Applied **RandomizedSearchCV** with **k-fold Cross-Validation** to identify the optimal set of hyperparameters for the AdaBoost model.
- Trained the final AdaBoost model using the **best parameter combination** on the **oversampled training dataset**.
- Evaluated and compared model performance metrics (Precision, Recall, F1-Score, and Accuracy) on both the **oversampled training data** and the **validation dataset** to assess generalization capability.

2. Random Forest Tuning with Undersampled Data

- Performed **hyperparameter optimization** using **RandomizedSearchCV** with **k-fold Cross-Validation** on the undersampled dataset.
- Trained the Random Forest model with the **best-fitting parameters** derived from the tuning process.
- Assessed model performance on both **undersampled training data** and the **validation dataset** to analyze bias-variance trade-offs.

3. Gradient Boosting Tuning with Oversampled Data

- Conducted **hyperparameter tuning** through **RandomizedSearchCV** coupled with **k-fold Cross-Validation** to determine the optimal parameter set for Gradient Boosting.
- Trained the Gradient Boosting model using the **oversampled training dataset** with the identified parameters.
- Compared key performance metrics between the **oversampled training data** and **validation dataset** to evaluate robustness and overfitting tendencies.

4. XGBoost Tuning with Oversampled Data

- Implemented **RandomizedSearchCV** with **k-fold Cross-Validation** to tune XGBoost hyperparameters efficiently.
- Re-trained the XGBoost model using the **optimized hyperparameters** on the **oversampled training dataset**.
- Measured and analyzed **performance metrics** across both **training** and **validation datasets** to ensure consistent and reliable predictive performance.

Solution Approach – Model Building and Selection

Continuation ----

Tuning AdaBoost using Oversampled data

- Select appropriate set of the parameters for **hyper tuning** and use **Randomized Search** techniques with **Cross-Validation (k-fold)** to find the optimal parameter combinations.
- Get best parameters fit and train Adaboost model with the oversampled trained data
- Observe the performance metrics with both Trained oversampled dataset and validation data set

Tuning Random forest using Undersampled data

- Select appropriate set of the parameters for **hyper tuning** and use **Randomized Search** techniques with **Cross-Validation (k-fold)** to find the optimal parameter combinations.
- Get best parameters fit and train Random Forest model with the Undersampled trained data
- Observe the the performance metrics with both Trained Undersampled dataset and validation data set

Tuning with Gradient boosting with Oversampled data

- Select appropriate set of the parameters for **hyper tuning** and use **Randomized Search** techniques with **Cross-Validation (k-fold)** to find the optimal parameter combinations.
- Get best parameters fit and train Gradient Boosting model with the oversampled trained data
- Observe the performance metrics with both Trained oversampled dataset and validation data set

Tuning XGBoost using Oversampled data

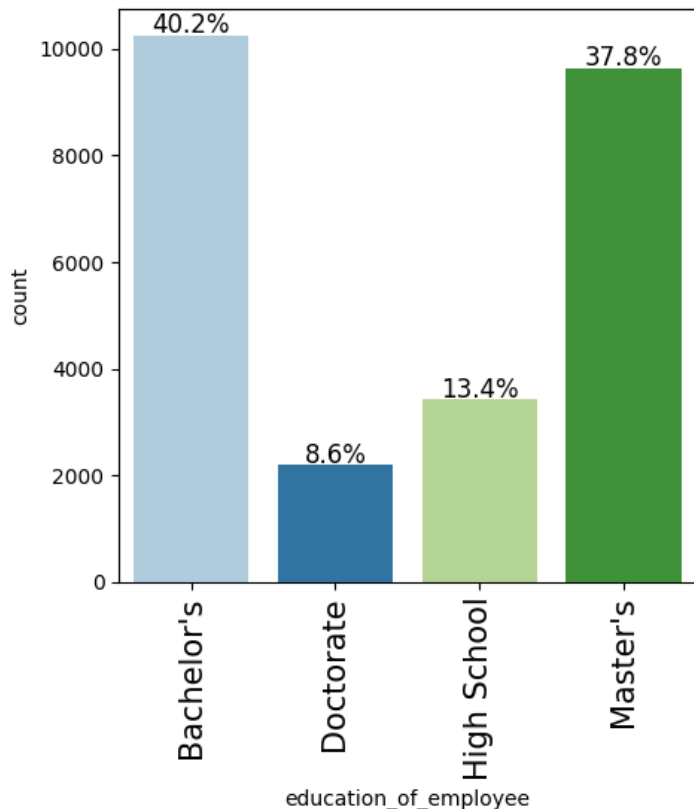
- Select appropriate set of the parameters for hyper tuning and use Randomized Search techniques with Cross-Validation (k-fold) to find the optimal parameter combinations.
- Get best parameters fit and train XGBoosting model with the oversampled trained data
- Observe the performance metrics with both Trained oversampled dataset and validation data

Final Model selection :

Perform a comprehensive **comparison and evaluation of all trained models** using the most appropriate performance metrics. The **final model will be selected based on the highest F1-score** and its ability to demonstrate strong **generalization performance** on unseen data.

This model **directly addresses the review burden** in alignment with the **OFLC's core objective** of improving efficiency and accuracy in application assessments.

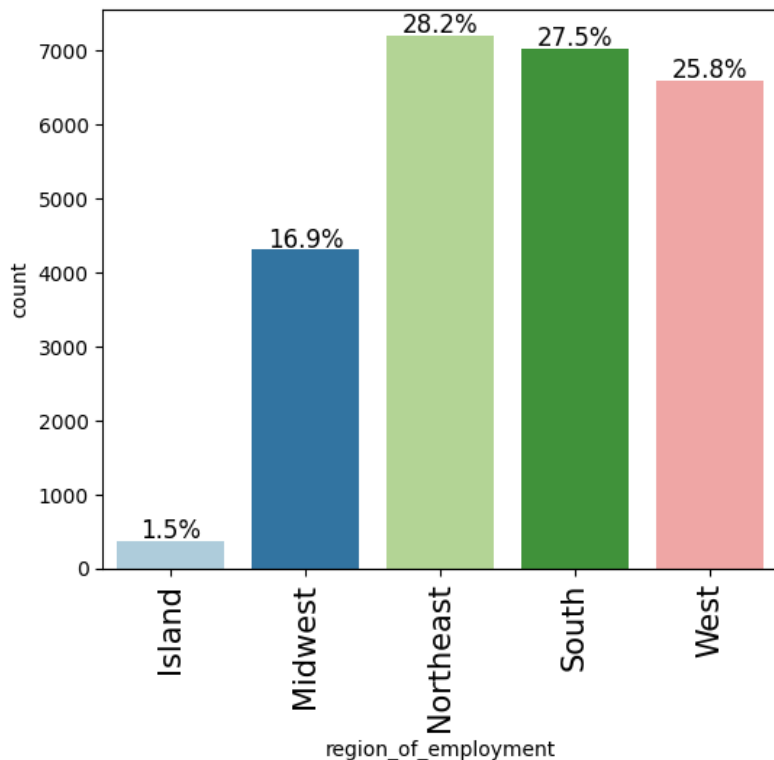
EDA Results- Univariate Analysis



Observations on education of employee

- **Bachelor's degrees are the most common education level** among employees, accounting for 40.2%.
- The combined percentage of employees with a Bachelor's or Master's degree is 78.0% (40.2%+37.8%).
- **Doctorate degrees are the least common education level**, representing only 8.6% of employees.
- There are more employees with a Master's degree (37.8%) than with a High School education (13.4%) and a Doctorate degree (8.6%) combined (22.0%).
- Employees with High School education (13.4%) are fewer than employees with either a Bachelor's (40.2%) or a Master's (37.8%) degree, but more than those with a Doctorate (8.6%)

EDA Results- Univariate Analysis



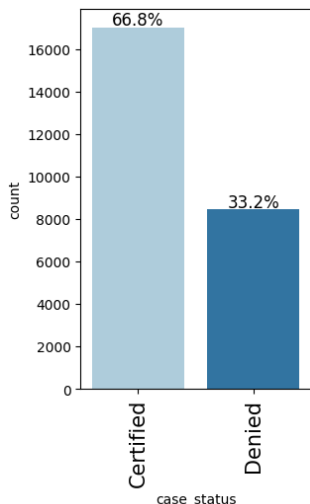
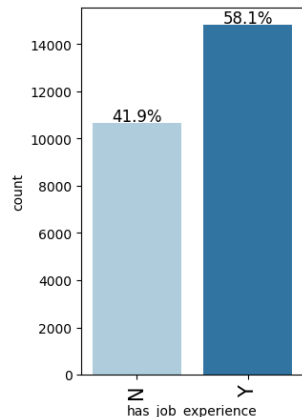
Observations on region of employment

- The **Northeast is the most common region** of employment, accounting for 28.2% of the total workforce.
- The combined percentage of employees in the Northeast (28.2%) and South (27.5%) regions is nearly 56% of the total.
- The **Island region represents a negligible portion** of the workforce, at only 1.5%.
- The Midwest region (16.9%) has significantly fewer employees than the Northeast, South, or West regions.
- The **distribution across the three largest regions (Northeast, South, and West) is relatively even, with percentages ranging narrowly from 25.8% to 28.2%..**

EDA Results- Univariate Analysis

Observations on Job Experience

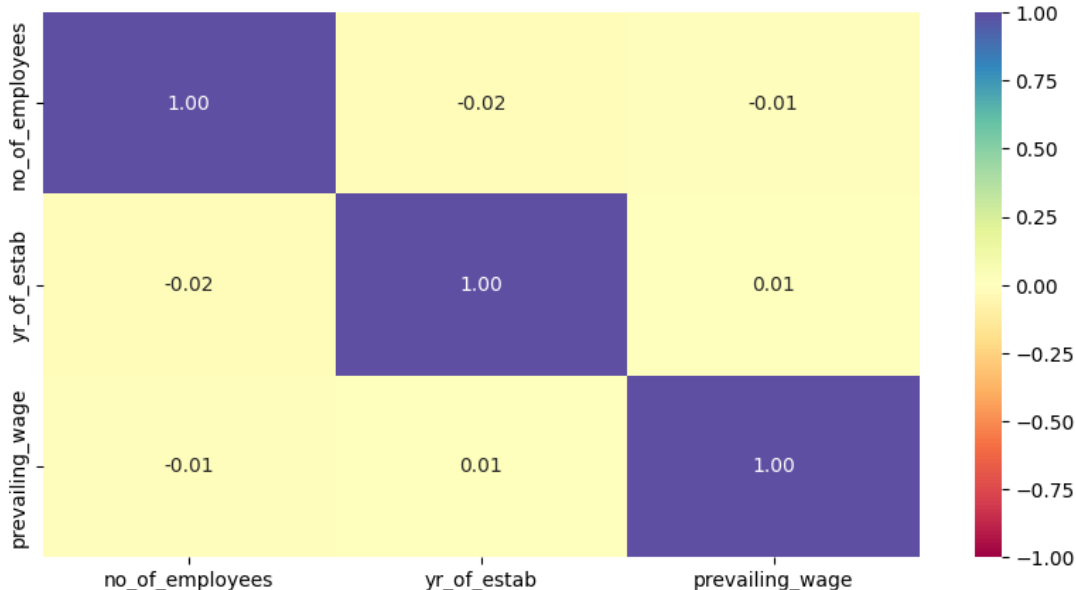
- A **majority of the individuals, 58.1%**, have job experience
- 41.9 % of the employee are not having job experience
- There are more than 14,000 individuals in the experienced group, while the **inexperienced group has over 10,000 individuals.**



Observations on the Case Status

- **Most of the employees(66.8%)** , cases are **certified**
- **33.2 % of the employee** cases are **denied**
- Certified cases are **approximately doubled** than the denied cases

EDA Results- Bivariate Analysis



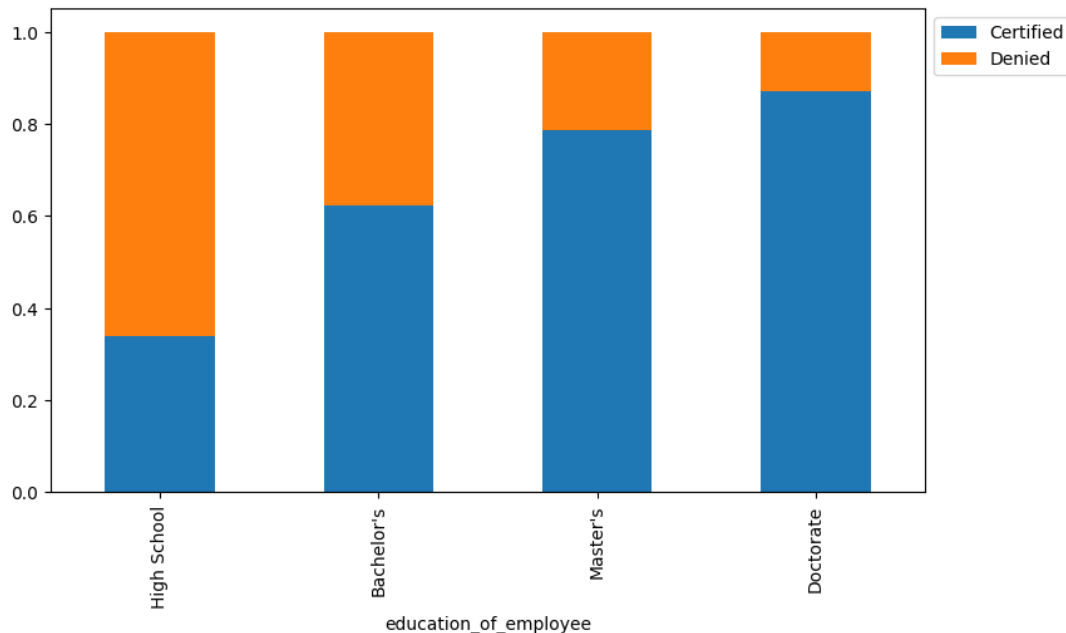
Observations on the Correlation Check

- **no_of_employees** and **yr_of_estab** have a negligible **negative correlation of -0.02**.
- The correlation between **no_of_employees** and **prevailing_wage** is **nearly zero at -0.01**.
- **yr_of_estab** and **prevailing_wage** are virtually **uncorrelated, showing a coefficient of only 0.01**.
- All three variables exhibit extremely **weak linear** relationships with each other, as all non-diagonal correlation values are very close to zero.
- There is **no multicollinearity concern** among these three numerical features since all correlation values are well below the typical threshold of 0.70.

[Link to Appendix slide on data background check](#)

EDA Results- Does higher education increase the chances of visa certification for well-paid jobs abroad?

Observations



- **Doctorate degree** holders have the **highest visa certification rate**, with close to **90%** of applications **certified**.
- The **High School** education category has the **lowest certification rate**, with only about **35%** of applications being certified.
- The trend shows a **clear, monotonic increase** in the certification rate as the level of education rises from High School to Doctorate.
- Applicants with a **Master's degree** have a certification rate of approximately **80%**, which is more than double the rate for High School graduates.
- A **Bachelor's degree significantly improves** certification chances to over **60%**, compared to the **35%** rate for High School.

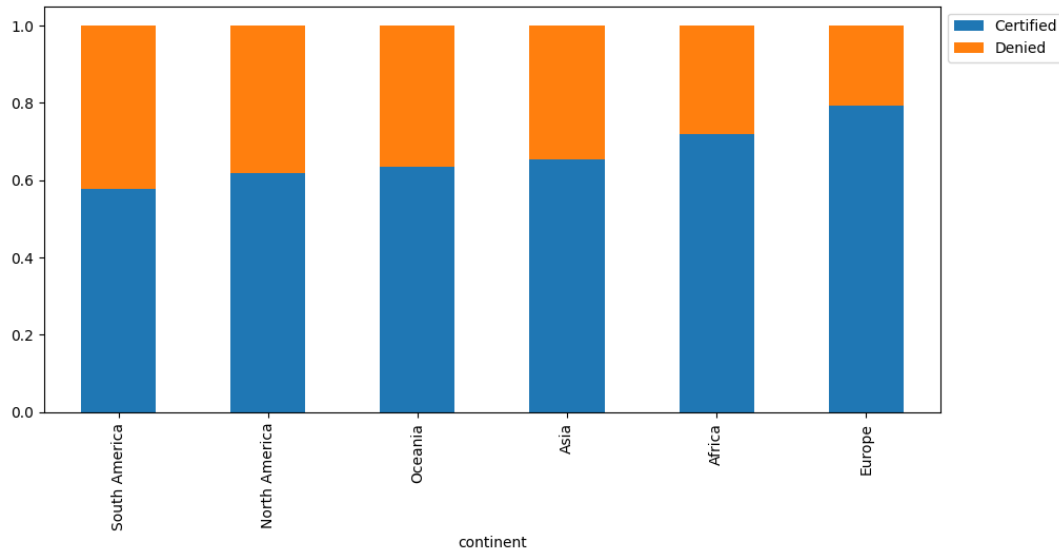
Conclusion:

Yes, Higher education increase the chances of visa certification for well-paid jobs abroad.

[Link to Appendix slide on data background check](#)

EDA Results- Does visa status vary across different continents?

Observations



- **Europe** has the **highest visa certification rate**, with approximately **80%** of applications being certified.
- **South America** has the **lowest visa certification rate**, with just under **60%** of applications certified.
- The certification rates for **South America, North America, Oceania, and Asia** are all clustered relatively closely, ranging from about **58% to 65%**.
- The combined group of Africa and Europe exhibits a markedly higher certification rate (both above 70%) compared to the four other continents.
- There is a significant **20 percentage point** difference in the certification rate between the **highest-ranking** continent (**Europe at ~80%**) and the **lowest-ranking** continent (**South America at ~58%**).

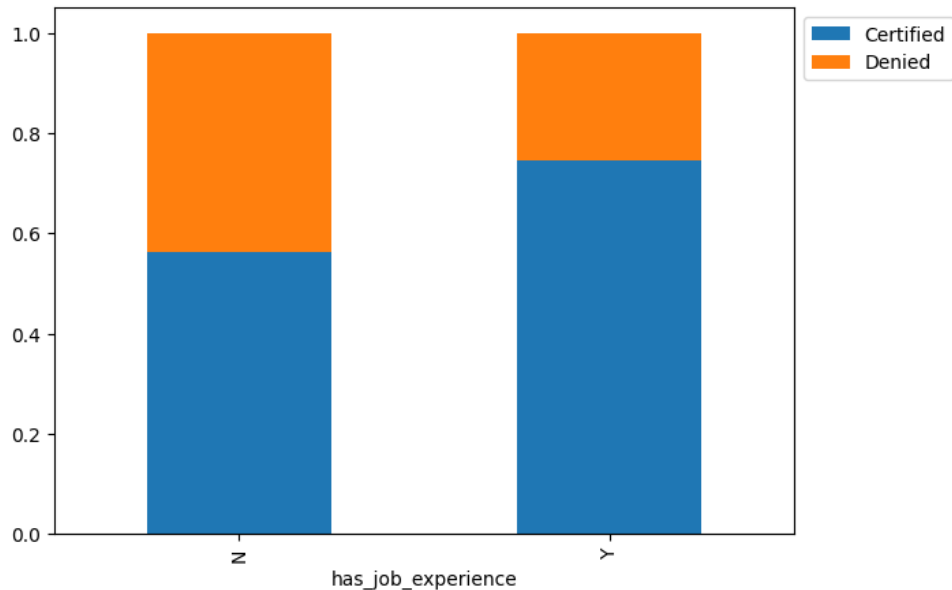
Conclusion:

Yes, Visa status **vary** across different continents.

[Link to Appendix slide on data background check](#)

EDA Results- Does having prior experience influence the Visa Approvals?

Observations



- **Having job experience** significantly **increases** the visa certification rate to **approximately 75%**.
- Applicants **without job experience** have a **much lower certification rate of approximately 55%**.
- The **denial rate for applicants** without job experience (45%) is nearly double the denial rate for those with job experience (25%).
- The presence of job experience acts as a **strong positive indicator**, increasing the **certification chance by about 20 percentage points (75%–55%)**.
- In both categories, the certification rate is above 50%, **but job experience provides a clear advantage in the application outcome**.

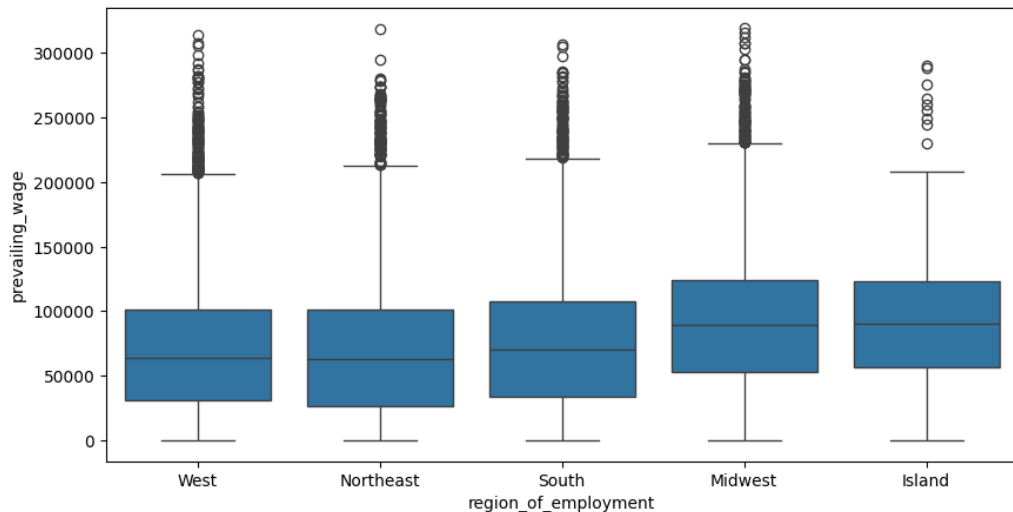
Conclusion:

Yes, having prior experience influence the Visa Approvals
And it will significantly **increase the chances of getting the Visa Approved**

EDA Results- Is the prevailing wage consistent across all regions of the US?

Observations

- The **Midwest region** exhibits the **highest median prevailing wage** (the line inside the box) compared to all other regions.
- The **Northeast region** has the **lowest median prevailing wage**, which is slightly less than the West and South regions.
- The **Island region** shows the **smallest interquartile range (IQR)**, indicating the **least variability** in the middle 50% of prevailing wages.
- **All regions**, particularly the **West and Northeast**, show a **large number of high outliers** in prevailing wage, extending well above \$200,000.
- The median prevailing wages for the **Midwest and Island regions** are **clearly higher (above \$100,000)** than those for the **West, Northeast, and South regions (below \$100,000)**.



Conclusion:

No, The prevailing wage is **not consistent across all regions** of the US

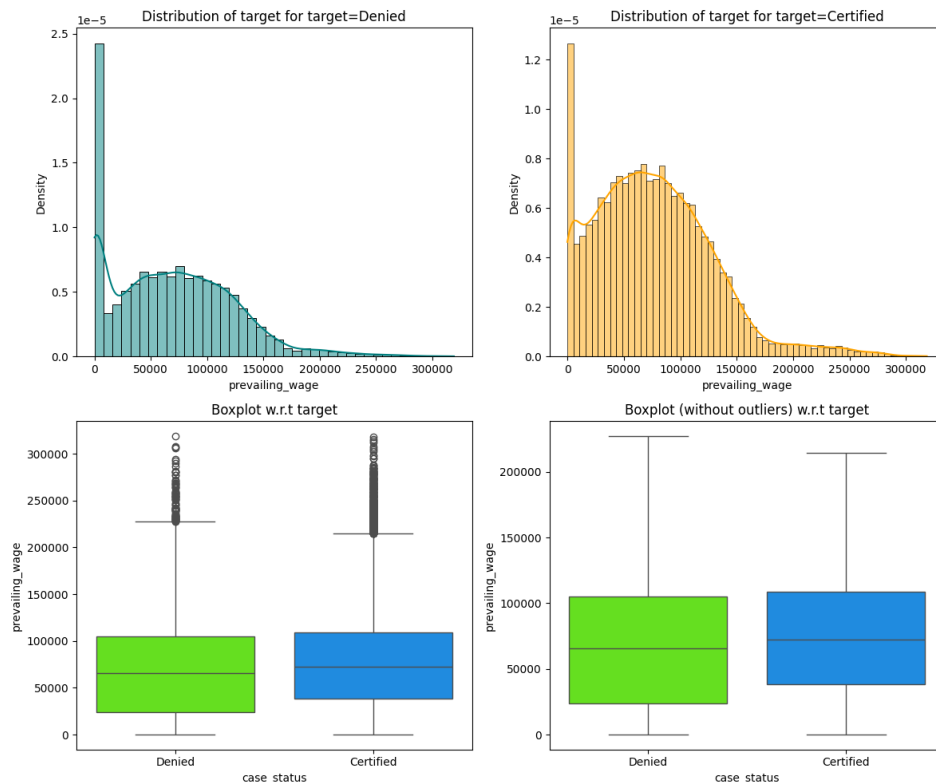
EDA Results- Does visa status vary with changes in the prevailing wage set to protect both local talent and foreign workers?

Observations

- **Median Wage is Higher for Certified Cases:** The median prevailing wage for Certified cases is visually higher than for Denied cases (as seen in both box plots).
- **Wage Distribution is Skewed Right for Both:** Both Certified and Denied distributions are **positively skewed**, with a long tail extending toward higher wages, though the Certified distribution is wider.
- **Low-Wage Applications Face High Denial Rates:** The distribution for Denied cases shows a **sharper peak** at the very low end of the wage spectrum (near zero), indicating a **high proportion of low-wage applications are denied**.
- **Interquartile Range is Similar:** The middle 50% of prevailing wages (the interquartile range or IQR) is roughly similar in size for both Denied and Certified cases.
- **High Outliers are Present in Both Categories:** Both Certified and Denied applications include numerous **high-wage outliers (above \$250,000)**, suggesting that a high wage does not guarantee certification.

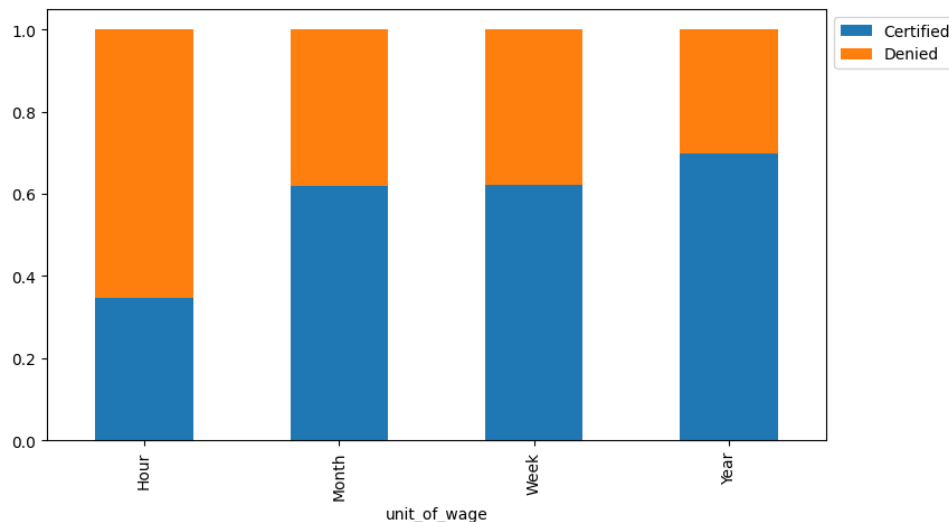
Conclusion:

Yes, The visa status (case status: Certified vs. Denied) does vary significantly with changes in the prevailing wage. The data clearly indicates a relationship between the prevailing wage and the outcome of the visa application



EDA Results- Does the unit of prevailing wage (Hourly, Weekly, etc.) have any impact on the likelihood of visa application certification?

Observations



- Wages reported on an **Annual (Year)** basis have the **highest visa certification rate** at approximately **70%**.
- Wages reported on an **Hourly (Hour)** basis have the **lowest visa certification rate** at approximately **35%**.
- The **denial rate for Hourly wages (~65%)** is nearly **double** the **denial rate for Annual wages (~30%)**.
- Wages reported on a **Monthly (Month) or Weekly (Week)** basis show a very similar and **moderate certification rate**, both **around 62%**.
- There is a **large ~35 percentage point difference in certification success** between the **Yearly unit of wage** and the **Hourly unit of wage**.

Conclusion:

Yes, the unit of prevailing wage (Hourly, Weekly, etc.) has a significant impact on the likelihood of visa application certification

Data Preprocessing

Duplicate Value Check

- Existing dataset **does not have a duplicate data**
- **Steps taken** : `data.duplicated()` is used to check if any duplicate data set exist

Missing Value Detection and Treatment (if needed with rationale)

- There is **No data missing** in the given dataset
- **Detection Steps**: `Data.isna().sum() = 0` for all columns

Data Cleaning

- Drop "**case_id**" as "**case_id**" is unique for each candidate and might not add value to modeling.
- **Fixed the negative values** in number of employees columns
 - Steps taken: In order to fix the **Absolute values** for number of employees is taken

Feature Engineering & Data preparation for Modeling

Encode categorical features.

- **Label Encoding** : `Case_Status` is label encoded **1 (positive class)** for “Certified” and **0 (Negative Class)** for “Denied” values
- **One-Hot Encoding of Features** : One Hot Encoding is done for other categorical columns by the method `pd.get_dummies` method and converted into the numerical column
- In order to prevent **multicollinearity**, dropped the first category of each variable by using `drop_first=True`

Data splitting

- 1st Split ((Train/Temporary Valid Split)) in the ratio of the 7:3
- 2nd Split (Final Validation and Test Set) ratio 9:1
 - 90% of the Xval data goes to the final validation set(27 % of the original DataSet)
 - 10% of the Xval data goes to the Test set(3 % of the original DataSet)

Overall Final Split Ratio (approximate):

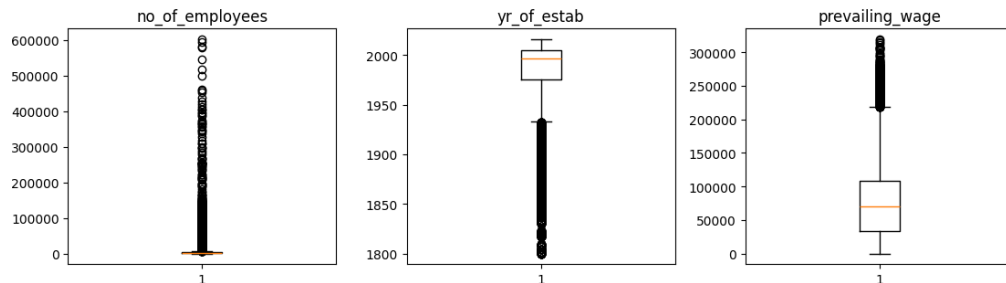
- **Train:** 70%
- **Validation:** 27%
- **Test:** 3%

Shape of the Split datasets

- ❖ **Training set** : (17836, 21)
- ❖ **Validation set** : (6879, 21)
- ❖ **Test set** : (765, 21)

Data Pre-Processing - Outlier Check

Steps Taken to detect the outlier : Outlier detection is done with the help of boxplot



Observations

- **no_of_employees:** This variable exhibits an **extreme number of high outliers**. The **interquartile range (IQR) is very small** (the box is close to the bottom axis), suggesting most companies have a **low number of employees**, while many scattered data points exist up to over **600,000 employees**.
- **yr_of_estab:** This variable shows **significant outliers on the low end**, with many establishment years falling far below the main cluster, reaching back to around the year 1800. The data is **highly skewed toward more recent establishment years (the box is near 2000)**.
- **prevailing_wage:** This variable contains a **large number of high outliers**, with many wages extending above the upper whisker, reaching up to over \$300,000. However, the outliers are less severe and less numerous compared to the no_of_employees column.

[Link to Appendix slide on data background check](#)

Defining scorer to be used for cross-validation and hyperparameter tuning

Best Metric Choice :

F1 –score is most balanced and practical choice here based on the problem statement and objectives of the OFLC to shortlist candidates with higher chances of visa approval. However, it's also important not to approve (predict positive) too many wrong cases to meet the Immigration and Nationality Act (INA) goal to protect the US workers against adverse impacts on their wages or working conditions by ensuring US employers' compliance with statutory requirements.

- It balances precision (avoiding false approvals)
- and recall (not missing genuine approvals).
- It's robust to class imbalance.
- It directly reflects the model's ability to make correct, confident predictions for the “visa certified” class.

Model Performance Summary and Final Model Selection

Trained Dataset - Metrics comparison for Hyper tuned models

Metrics	Gradient Boosting tuned with oversampled data	XGBoost tuned with oversampled data	AdaBoost tuned with oversampled data	Random forest tuned with under sampled data
Accuracy	0.8020649710400403	0.8067657181230589	0.7875010492739024	0.7494512915752153
F1	0.8133164436703348	0.8176489226869454	0.7981823255072348	0.7561216105176664
Precision	0.7695707543636228	0.7740532433445819	0.7599817822984667	0.7365135264927165
Recall	0.8623352639973139	0.8664484176949551	0.8404264249139596	0.776802296133716

Validation Dataset- Metrics comparison for Hyper tuned models

Metrics	Gradient Boosting tuned with oversampled data	XGBoost tuned with oversampled data	AdaBoost tuned with oversampled data	Random forest tuned with under sampled data
Accuracy	0.7399331298153802	0.7361535106846926	0.7345544410524786	0.7123128361680476
F1	0.8141684844707593	0.8113893796113478	0.8085953878406709	0.7734401831711506
Precision	0.7786608384661236	0.7762974746470471	0.7798220784472301	0.8157449891330596
Recall	0.8530692207226818	0.8498040922942969	0.8395733565520244	0.7353069220722682

Key Observations

- Based on above comparison, **Gradient Boosting tuned with oversampled data** model has demonstrated
 - High and consistent F1 Score across** Testing DataSets (0.81133) and validation Dataset(0.8141)
 - Strong balance**, with a **Recall of 0.8530** (highest of all models) and **Precision of 0.7766** on the **validation set**
- While the XGBoost model achieved a high F1-Score of 0.8176 on the training data, but a **slight performance degradation to 0.8113** on the validation set suggests a **minor degree of overfitting**

Model Performance Summary and Final Model Selection

Final Model : The Gradient Boosting tuned with oversampled

Model	F1 Score (Trained Dataset)	F1 Score (Validation Dataset)	Difference (Trained - Validation)
Gradient Boosting	0.8133	0.8141	-0.0008
XGBoost	0.8176	0.8113	0.0063
AdaBoost	0.7981	0.8085	-0.0104
Random Forest	0.7561	0.7734	-0.0173

Justification and Analysis

- ✓ **Highest Validation F1 Score:** The Gradient Boosting tuned with oversampled data model achieves the highest F1 Score on the crucial Validation Dataset at 0.8141. Since the validation set is the best proxy for unseen data, its performance here is paramount.
- ✓ **Excellent Generalization (Minimal Overfitting):** This model exhibits a rare and highly desirable trait: its F1 Score on the Validation Dataset (0.8141) is slightly higher than its F1 Score on the Trained Dataset (0.8133). This indicates exceptional generalization and virtually no overfitting.
- ✓ **Strong Precision and Recall:** While all models perform similarly on these metrics, Gradient Boosting maintains a strong balance, with a Recall of 0.8530 (highest of all models) and Precision of 0.7766 on the validation set, which collectively contribute to the highest F1 score.

Model Performance Summary a Final Model –Test DataSets

Final Model :The Gradient Boosting tuned with oversampled

Accuracy	Recall	Precision	F1-Score
0.7581699346405228	0.87279843444227	0.7879858657243817	0.8282265552460538

Key Observations

- The model achieves a **final F1-Score of approximately 0.8282**, demonstrating a strong balance between Precision and Recall.
- The overall Accuracy of the model is approximately 75.82%, meaning it correctly classifies about three out of four visa cases.
- The **Recall is very high at approximately 0.8728**, indicating the model is highly effective at identifying the true positive visa approvals.
- The Precision is approximately 0.7880, suggesting that **roughly 78.8% of the cases predicted as approved were correctly certified**.
- The Recall (0.8728) is the highest metric, **being approximately 9.5 percentage points higher than the Precision (0.7880)**.

Model Performance Summary Final Model

Final Model :

Gradient Boosting tuned with oversampled

DataSets	F1-Score
Training DataSets	0.8133164436703348
Validation DataSets	0.8141684844707593
Testing DataSets	0.8282265552460538

Best Parameters obtained after Hyper tuning and used in model :

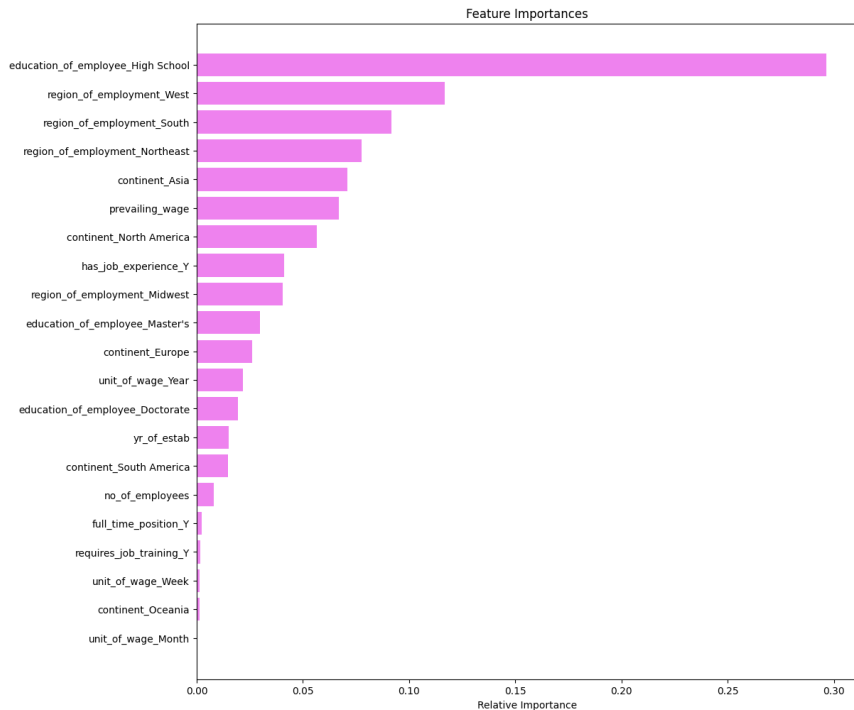
1. n_estimators : 200
2. learning_rate: 0.1,
3. Subsample :0.9999999999999999
4. max_features :0.4
5. CV score=0.8022045072757036

Key Points

- ✓ **High-Confidence Predictive Performance:** The model has demonstrated a **high and consistent F1 Score across all evaluation sets, culminating in an 82.8% F1 Score on the unseen Testing Data.** This metric confirms the model's robust ability to accurately classify visa applications, ensuring a strong balance between identifying approvable cases (Recall) and minimizing false approvals (Precision).
- ✓ **Exceptional Generalization and Reliability:** The model exhibits **zero evidence of overfitting**, as shown by the F1 Score being slightly higher on the Validation (**0.8141**) and Testing (**0.8282**) datasets compared to the Training set (**0.8133**). This guarantees the solution will perform reliably and predictably when deployed to screen live, new applications.
- ✓ **Directly Addresses Review Burden (OFLC Objective):** Implementing this solution will **facilitate the visa approval process** by automatically flagging applicants with a high probability of certification. This allows OFLC reviewers to focus their time and resources on complex or borderline cases, significantly reducing the manual review burden caused by the increasing application volume.
- ✓ **Data-Driven Recommendation System:** The classification model provides **the foundation to recommend a suitable profile for certification or denial by analyzing the most influential drivers (features) in the data.** This moves the decision process from subjective review to an objective, data-driven system, ensuring consistent and fair outcomes.

Model Performance Summary and Final Model Selection

Feature Importance :



Key Observations

- **Employee Education is the Single Most Dominant Factor:** The feature `education_of_employee_High School` is the overwhelmingly **most important predictor**, contributing nearly **30%** of the relative importance and significantly outweighing all other features.
- **Geographic Location is a Critical Driver:** The region of employment is **highly influential**, with three specific regions—**West, South, and Northeast**—ranking among the **top five most important features overall**.
- **Specific Continents are More Important than Prevailing Wage:** The specific continent of origin, particularly **continent_Asia**, is a **stronger predictor** than the general `prevailing_wage` of the job, which suggests that applicant origin plays a crucial role beyond simple salary metrics.
- **Master's and Doctorate Degrees Have Low Importance:** Higher educational levels like `education_of_employee_Master's` and `education_of_employee_Doctorate` have very low relative importance compared to a **High School education**, suggesting **having an advanced degree is not a primary driver for visa certification in this model**.

APPENDIX

Data Background and Contents

- The Existing dataset has **25480 rows and 12 columns**
- Datatypes of the columns
 - float64(1)
 - int64(2)
 - object(9)
- Data Duplicate Check
 - Existing Dataset is not having duplicate data
- Missing value Check
 - Existing dataset is not having any missing value

Data Background and Contents-Statstical Summary

- The mean number of employees (~5,667) is much larger than the median (~2,109), indicating a **strong positive skew and the presence of high outliers**.
- The maximum number of employees (~602,069) **is an extreme outlier**, pulling the **standard deviation (22,878)** significantly high.
- The no_of_employees feature contains **impossible minimum value (-26)**, suggesting the presence **of data quality issues (negative values)**.
- The interquartile range (IQR) for the **yr_of_estab** is **relatively** small, with 50% of establishments founded between 1976 and 2005.
- The prevailing_wage has a mean (\$74,456) slightly higher than its median (\$70,308), **indicating a mild positive skew with high outliers up to \$319,210**.

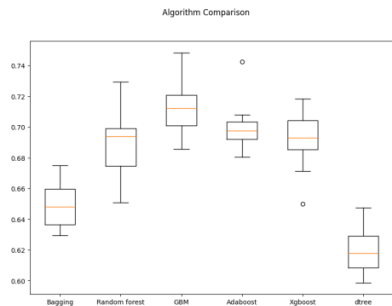
Model Building - Original Data

List of the model built and F1 Scores

Index	Model	F1-Score Test Datasets	F1-Score Validation Datasets
1	Bagging	0.7754467419381872	0.7675817565350541
2	Random Forest	0.8031383296203435	0.7972364702187794
3	GBM	0.8236772733014497	0.8195818459969403
4	Adaboost	0.8195228965788536	0.8158053488839735
5	XGBoost	0.8071328600156471	0.8083918974782968
6	DecesionTree	0.7398268032422285	0.7477497255762898

Key Insights:

- **GBM (Gradient Boosting Machine)** is the **best-performing model** overall, achieving the highest F1-Scores on both the Test (0.8237) and Validation (0.8196) sets.
- The models **GBM, Adaboost, and Random Forest** form a high-performance cluster, all scoring above 0.81 on the Test set.
- The **Decision Tree model** is the **worst-performing model** with the lowest F1-Scores, suggesting it suffers from high bias or underfitting compared to the ensemble methods.
- All models show **low variance/good generalization**, as the F1-Scores on the Test and Validation sets are very close, **with no signs of severe overfitting**.
- **Bagging** exhibits the **largest performance drop (~1%)** when moving from the Validation set to the Test set, though its generalization is still acceptable.



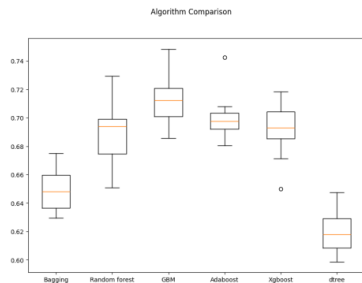
Model Building - Oversampled Data

Oversampling technique : SMOTE

- Before OverSampling, counts of label '1': 11913
- Before OverSampling, counts of label '0': 5923
- ❖ After OverSampling, counts of label '1': 11913
- ❖ After OverSampling, counts of label '0': 11913

List of the model built and F1 Scores

Index	Model	F1-Score Test Datasets	F1-Score Validation Datasets
1	Bagging	0.7597364450891436	0.7606724176067242
2	Random Forest	0.7949888607780384	0.7953896584540552
3	GBM	0.8061052289956541	0.8125259228535877
4	Adaboost	0.8004129012242016	0.8120255086547221
5	XGBoost	0.8016497215584085	0.8039950062421972
6	DecesionTree	0.7260765028494454	0.7387687188019967



Key Insights:

- **GBM (Gradient Boosting Machine)** is the **top-performing model with the highest F1-Score (0.8061)** on the crucial Test dataset.
- The models **GBM, XGBoost, and Adaboost** form the high-performance tier, all achieving an F1-Score of 0.80 or higher on the Test set.
- The **Decision Tree model is significantly the worst performer**, scoring more than 7 points lower (0.7261) than the leading models.
- **All ensemble models (Bagging, Random Forest, Boosting) demonstrate strong generalization**, as their Test F1-Scores are within a **very narrow range of their Validation F1-Scores**.
- **Adaboost and GBM exhibit a slight ~1% performance drop on the Test set compared to the Validation set**, indicating the **XGBoost model generalizes marginally better to unseen data..**

Model Building - Undersampled Data

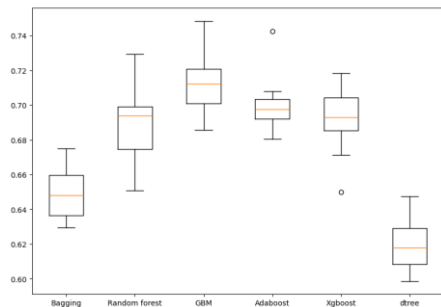
Undersampling technique : `RandomUnderSampler` as a type of under-sampling technique that works by reducing the number of instances in the majority classes to balance the class distribution of the training data is used here

- Before UnderSampling, counts of label '1': 11913
- Before UnderSampling, counts of label '0': 5923
- ❖ After UnderSampling, counts of label '1': 5923
- ❖ After UnderSampling, counts of label '0': 5923

List of the models built and F1 Scores

Index	Model	F1-Score Test Datasets	F1-Score Validation Datasets
1	Bagging	0.648376671972561	0.6916956737941323
2	Random Forest	0.6875023869467406	0.734144015259895
3	GBM	0.712085878138409	0.7608695652173914
4	Adaboost	0.7004337719805565	0.7604202747950584
5	XGBoost	0.6916133208114543	0.7423652871123688
6	DecesionTree	0.620256427342062	0.6839080459770115

Algorithm Comparison



Key Insights:

- **Ensemble Methods (Boosting) Consistently Dominate: In all three scenarios**, the boosting algorithms (GBM, Adaboost, and XGBoost) rank in the top three, confirming their general superiority in predictive power for this classification task.
- **GBM Exhibits the Best and Most Stable Generalization:** GBM is the overall best-performing model (**highest Test F1-Score**) in two of the three tables (image_05d40f and image_065b0c) and holds a consistent lead or very close second in all scenarios.
- **Decision Tree is Universally the Weakest Performer:** The Decision Tree consistently records the lowest F1-Score on the Test dataset in all three scenarios, underscoring the necessity of using ensemble techniques for robust performance.
- **Boosting Methods Handle Scenario Variation Better than Bagging:** Across all three performance shifts, the boosting models (**GBM, Adaboost, XGBoost**) maintain a tighter F1-Score range (and better relative rank) compared to the wider swings seen in the Bagging and Random Forest models.

Model Improvement - Hyperparameter Tuning

Trained Dataset - Metrics comparison for Hyper tuned models

Metrics	Gradient Boosting tuned with oversampled data	XGBoost tuned with oversampled data	AdaBoost tuned with oversampled data	Random forest tuned with under sampled data
Accuracy	0.8020649710400403	0.8067657181230589	0.7875010492739024	0.7494512915752153
F1	0.8133164436703348	0.8176489226869454	0.7981823255072348	0.7561216105176664
Precision	0.7695707543636228	0.7740532433445819	0.7599817822984667	0.7365135264927165
Recall	0.8623352639973139	0.8664484176949551	0.8404264249139596	0.776802296133716

Validation Dataset- Metrics comparison for Hyper tuned models

Metrics	Gradient Boosting tuned with oversampled data	XGBoost tuned with oversampled data	AdaBoost tuned with oversampled data	Random forest tuned with under sampled data
Accuracy	0.7399331298153802	0.7361535106846926	0.7345544410524786	0.7123128361680476
F1	0.8141684844707593	0.8113893796113478	0.8085953878406709	0.7734401831711506
Precision	0.7786608384661236	0.7762974746470471	0.7798220784472301	0.8157449891330596
Recall	0.8530692207226818	0.8498040922942969	0.8395733565520244	0.7353069220722682

Key Observations

- Based on above comparison, **Gradient Boosting tuned with oversampled data** model has demonstrated
 - High and consistent F1 Score across** Testing DataSets (0.81133) and validation Dataset(0.8141)
 - Strong balance**, with a **Recall of 0.8530** (highest of all models) and **Precision of 0.7766** on the **validation set**
- While the XGBoost model achieved a high F1-Score of 0.8176 on the training data, but a slight performance degradation to 0.8113 on the validation set suggests a minor degree of overfitting

Model Improvement - Hyperparameter Tuning

Method used to Hyper tuned for all models:

RandomizedSearchCV

AdaBoost using Oversampled data-Parameter Tuned

1. `n_estimators`: [50,75,100,125,150],
2. `learning_rate`: [1.0,0.5,0.1,0.01],
3. `estimator`: [DecisionTreeClassifier(max_depth=1, random_state=1), DecisionTreeClassifier(max_depth=2, random_state=1), DecisionTreeClassifier(max_depth=3, random_state=1),

Best Parameters obtained :

1. `n_estimators` : 100
2. `learning_rate`: 1.0
3. CV score= 0.7983888515430675
4. Estimator : DecisionTreeClassifier(max_depth=2, random_state=1)

Random forest using undersampled data-Parameter Tuned

1. `"n_estimators"`: [50,75,100,125,150]
2. `"min_samples_leaf"`: [1,2,4,5,10]
3. `"max_features"`: np.arange(0.2,0.8,0.1)
4. `"max_samples"`: np.arange(0.5,1.0,0.1)

Best Parameters obtained :

1. `n_estimators` : 125
2. `min_samples_leaf`: 10,
3. `max_samples` :0.7
4. `max_features` :0.2
5. CV score= 0.7219429413158686

Model Improvement - Hyperparameter Tuning

Method used to Hyper tuned for all models:
RandomizedSearchCV

Gradient Boosting using Oversampled data-Parameter Tuned

1. "n_estimators": [100,125,150,175,200,225,250]
2. "learning_rate": [0.1, 0.05, 0.01, 0.005]
3. "subsample": np.arange(0.7,1.0,0.1)
4. "max_features": [0.3,0.4,0.5,0.6]

Best Parameters obtained :

1. n_estimators : 200
2. learning_rate: 0.1,
3. Subsample :0.9999999999999999
4. max_features :0.4
5. CV score=0.8022045072757036

XG Boosting using Oversampled data-Parameter Tuned

1. n_estimators': [50,75,100,125,150,175,200,225,250],
2. 'subsample': np.arange(0.5,1.0,0.1),
3. 'gamma': [1,3,5,8],
4. 'colsample_bytree': [0.3,0.5,0.7,1.0],
5. 'colsample_bylevel': [0.3,0.5,0.7,1.0]

Best Parameters obtained :

1. n_estimators : 150
2. gamma: 3,
3. Subsample :0.7
4. colsample_bytree :1.0
5. colsample_byleve:0.3
6. CV score=0. 8031667550158321



Happy Learning !

